

# Synthetic Aperture Local Conformal Autoencoder for Semi-Supervised Speaker's DOA Tracking

Idan Cohen , Sharon Gannot , *Fellow, IEEE*, and Ofir Lindenbaum 

**Abstract**—In this article, we address the problem of tracking the direction of arrival (DOA) of a moving speaker in noisy and reverberant environments. We aim to achieve this in real-time, using as few measurements as possible. Toward this goal, we present a semi-supervised learning scheme that tracks the speaker using acoustic features only. Specifically, we extend a recently proposed method called local conformal autoencoder (LOCA), a deep neural network (DNN)-based dimensionality reduction technique that considers the local information from adjacent measurements. We design a unique training procedure for LOCA, which uses a synthetic aperture training paradigm to take advantage of the speaker's movement during training. We also add an anchoring loss term to the unsupervised LOCA model, which improves training stability and establishes a connection between the encoder mapping and the real-world position of the speaker. Finally, we conduct a comprehensive simulation study to demonstrate the effectiveness of our proposed method in dynamic environments with various levels of noise and reverberation.

**Index Terms**—Semi-supervised learning, speaker tracking, synthetic aperture, LOCA, RTF.

## I. INTRODUCTION

**S**PEAKER DOA estimation and, in particular, speaker tracking are essential tasks in numerous domains, ranging from automated conferencing solutions [1], [2] to advanced surveillance systems [3], [4]. With precise speaker tracking, these technologies can enhance sound capture quality, facilitate more efficient communication, and improve security measures.

Classical solutions for this problem, namely the generalized cross-correlation phase transform (GCC-PHAT) [5] and its multichannel extension, the steered-response power phase transform (SRP-PHAT) [6], rely on analyzing the cross-correlation between the signals and extracting their peak, which corresponds with the time difference of arrival (TDOA). Other methods, such as the multiple signal classification (MUSIC) algorithm [7] and its extensions, offer high-resolution capabilities and rely on analyzing the covariance matrix of the received signals. These techniques, initially designated to address single-path static scenarios, demonstrate limitations when dealing with the dynamic

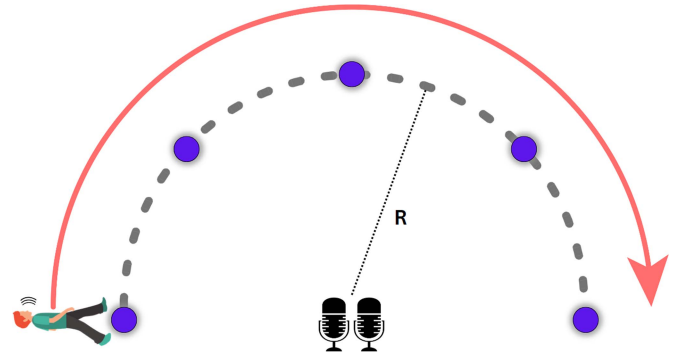


Fig. 1. An illustration of the experimental setup. Dynamic speakers are confined to an arc of radius  $R$  around a pair of microphones, both in the training and test phases.  $0 \leq \theta(t) \leq 180$ . Blue circles denote labeled samples.

multi-path case. In [8] and [9], a grid search was conducted over the output power of a beamformer steered to all potential locations in order to estimate the position. Several recursive least squares-based tracking methods were presented in [10], and probabilistic approaches were presented in [11], [12], [13].

In a prevalent probabilistic approach for speaker tracking, the source path is modeled as a stochastic process, enabling the utilization of Bayesian inference frameworks. In this approach, the TDOAs of the sources are estimated, resulting in nonlinear state-space model equations. Multiple algorithms were proposed in this context, including the Kalman filter [14], [15] and its nonlinear extensions, specifically the extended Kalman filter (EKF) [16], [17], the unscented Kalman filter (UKF) [16], the multiple model Kalman filter (MMKF) [18], and the wrapped Kalman filter (WKF) [19]. Particle filters were also suggested as a viable solution [20], [21], [22], [23], [24], together with hidden Markov models (HMMs) [25], [26], probability hypothesis density (PHD) filters [27], [28], [29], [30] and factor graphs [31]. Similarly, non-Bayesian approaches, based on the recursive expectation-maximization (REM) procedure, were suggested at [32], [33], [34], [35], [36], [37].

Recently, greater focus has been directed towards training-based localization methods. These approaches are usually divided into two groups—supervised and semi/weakly-supervised. In the supervised schemes, the training dataset consists of samples whose positions are accurately known. While achieving excellent performance, the supervised approaches require extensive amounts of labeled data, which are often laborious to collect. Such approaches include [38] and [39], which suggest applying convolutional neural networks (CNNs)

Received 20 September 2024; revised 8 April 2025 and 21 June 2025; accepted 26 June 2025. Date of publication 14 July 2025; date of current version 18 July 2025. The work was supported by a grant from the Audition Project, Data Science Program, Council of Higher Education, Israel. The associate editor coordinating the review of this article and approving it for publication was Dr. Romain Serizel. (Corresponding author: Sharon Gannot.)

The authors are with Bar-Ilan University, Ramat-Gan 5290002, Israel (e-mail: idan.cohen@biu.ac.il; sharon.gannot@biu.ac.il; ofir.lindenbaum@biu.ac.il).

Digital Object Identifier 10.1109/TASLPRO.2025.3587465

and fully connected networks (FCNs) to the acoustic features, respectively. In [40], the authors suggest improving the former approach by integrating a long short-term memory (LSTM) network to utilize the temporal relations between consecutive time frames. Other works, such as [41] and [42], suggest using a convolutional recurrent neural network (CRNN). In this case, the recurrent layers model the evolution of the spatial parameters, given the sequential features and their spatial trajectory. In [43], the authors suggest applying a supervised U-Net structure with a dense block to denoise the GCC-PHAT components before constructing the SRP-PHAT, achieving improved tracking. In [44], a narrow-band variant of the SRP-PHAT was used to train a deep time-frequency-based classification model. The proposed approach allows for the localization of multiple concurrent speakers while circumventing the challenge of a possible mismatch between microphone array configurations in the training and test stages.

In the second group, which includes semi/weakly supervised approaches, a deeper understanding of the nature of the data is required to exploit the labeled information efficiently. In [45] and [46], a graph-Laplacian-based manifold learning approach is used to learn the latent manifold that controls the acoustic data. In [47], a variational auto encoder (VAE) is used to learn the probability distribution that characterizes the acoustic data. In [48], a weakly-supervised setting is considered for localization, where additional information regarding the spatial relationship of samples is given in the form of triplets. The triplets consist of an anchor, a positive, and a negative sample—based on their proximity to the anchor. This work was later extended to the dynamic case in [49]. A comprehensive survey of sound source localization with deep learning methods can be found in [50].

In our paper, we adopt the recently proposed LOCA framework [51] for the task of semi-supervised speaker tracking. LOCA is a manifold learning-based approach that was proved viable for the task of 2-dimensional acoustic scene mapping [52]. At the core of the approach lies an auto-encoder, which aims to extract a low-dimensional latent embedding that is isometric to the true latent manifold that underlies the data. In Section II, we formalize the problem and present the instantaneous acoustic features used for tracking. In Section III, we introduce new extensions for LOCA, which aim to take advantage of the speakers' movement during training. Our solution involves adopting the synthetic aperture training paradigm. Additionally, we adopt a few methodological modifications to LOCA's training framework, aiming to achieve a more stable convergence. Specifically, we modify the minimization process and introduce the anchoring loss term to align the learned embedding to the real-world coordinates efficiently. In Section IV, we benchmark our proposed scheme against previously discussed semi/weakly supervised approaches and evaluate them through a series of simulations. We also discuss the limitations of our approach, focusing on its performance under various noisy and reverberant conditions. Additionally, we analyze several other factors to determine their influence on the tracking performance. The discussion is followed by an ablation study to assess the impact of the proposed methodological changes.

## II. PROBLEM FORMULATION

In this section, we formalize the problem and introduce the instantaneous relative transfer functions (iRTFs)—the acoustic features used to implement the acoustic tracking.

### A. Problem Statement

Consider an acoustic enclosure where a speaker is moving and recorded by a pair of microphones. Following previous works [46], [49], we focus on the one-dimensional setting, where the speaker can only travel on an arc of radius  $R$  around a microphone pair. In this setting, the DOA, denoted  $\theta(t)$ , represents the time-varying angular displacement of the speaker along the arc path around the microphone pair. We also assume that we are given a small set of anchor points, i.e., samples whose positions are known and labeled. Importantly, in our one-dimensional framework, “position” and “location” always refer to the DOA rather than Cartesian coordinates. Hence, spatial information throughout this work is understood in terms of angular displacement rather than absolute spatial positioning. The discussed setting is illustrated in Fig. 1.

### B. Acoustic Features - iRTF

We consider time-domain acoustic measurements of the form

$$z_i(t) = \{a_i * s\}(t) + u_i(t), \quad (1)$$

with  $s(t)$  being the source signal,  $a_i(t)$ , the unknown room impulse responses (RIRs) relating the source and each of the microphones,  $i = \{1, 2\}$ , and  $u_i(t)$  the noise signals independent of the source.

It follows that in the time-frequency domain, we can approximate the measured signals as:

$$z_i(\ell, k) \approx a_i(\ell, k) s_i(\ell, k) + u_i(\ell, k), \quad (2)$$

where  $s_i(\ell, k)$ ,  $a_i(\ell, k)$ , and  $u_i(\ell, k)$  are the time-frequency domain representations of  $s_i(t)$ ,  $a_i(t)$  and  $u_i(t)$ , respectively, at time frame index  $\ell$  and frequency index  $k$ .

Note that, in our case, the speakers are moving, and we, therefore, have to work with instantaneous acoustic features, namely the instantaneous relative transfer functions (iRTFs). We select  $z_1(n)$  as the reference signal and define the iRTF as:

$$h(\ell, k) = \frac{a_2(\ell, k)}{a_1(\ell, k)}. \quad (3)$$

We will use a set of  $N_c$  context time frames to estimate the iRTF at any given time frame index  $\ell$ . Without loss of generality, in our formulation, the parameter  $N_c$  is assigned even values. Assuming negligible noise, the iRTF,  $\hat{h}(\ell, k)$ , can be estimated from:

$$\hat{h}(\ell, k) \approx \frac{\hat{\phi}_{z_2 z_1}(\ell, k)}{\hat{\phi}_{z_1 z_1}(\ell, k)} \approx \frac{\sum_{\ell'=\ell-\frac{N_c}{2}}^{\ell+\frac{N_c}{2}} z_2(\ell', k) z_1^*(\ell', k)}{\sum_{\ell'=\ell-\frac{N_c}{2}}^{\ell+\frac{N_c}{2}} z_1(\ell', k) z_1^*(\ell', k)}, \quad (4)$$

where  $\hat{\phi}_{z_1 z_1}(\ell, k)$  and  $\hat{\phi}_{z_2 z_1}(\ell, k)$  are the power spectral density (PSD) and the cross-PSD at time frame  $\ell$ , respectively. Ultimately, we focus on a subset of the frequency bins range,

namely  $k_1, \dots, k_P$ , that includes most spectral power. This would allow for a reliable estimation of the acoustic features, further improving the localization accuracy. We finally define our feature vector at time frame  $\ell$  as the concatenation of the real-value and imaginary-value components of the iRTFs in the designated frequency range:

$$\hat{h}(\ell) = \begin{bmatrix} \text{Re} \left( \hat{h}(\ell, k_1), \dots, \hat{h}(\ell, k_P) \right), \\ \text{Im} \left( \hat{h}(\ell, k_1), \dots, \hat{h}(\ell, k_P) \right) \end{bmatrix}^\top. \quad (5)$$

### III. PROPOSED TRACKING ALGORITHM

#### A. The Acoustic Manifold

Previous works have studied the nature of the relative transfer function (RTF) acoustic features [53], [54], [55]. RTFs are high-dimensional representations in  $\mathbb{C}^D$  that correspond to the vast amount of reflections from different surfaces characterizing the enclosure. Essentially, the RTFs depict complex acoustic reflection patterns that serve as ‘fingerprints,’ uniquely distinguishing the source position in the enclosure. Following the manifold hypothesis [56], we assume that the RTF features are not spread uniformly in the entire space of  $\mathbb{C}^D$ . Instead, they are confined to a compact nonlinear manifold  $\mathcal{M}$  of dimension  $d$ , much smaller than the ambient space dimension, i.e.,  $d \ll D$ . This assumption is justified by the fact that perturbations of the RTF are only influenced by a small set of parameters related to the physical characteristics of the environment. These parameters include the enclosure dimensions, its shape, the surfaces’ materials, and the positions of the microphones and the source. Moreover, we focus on a static configuration, where the enclosure properties and the microphone array position remain fixed. In such an acoustic environment, the only varying degree of freedom is the location of the source.

Although nonlinear, in small neighborhoods, the manifold is locally linear, meaning that it is flat in the vicinity of each point and coincides with the tangent plane to the manifold at that point [55]. Hence, the Euclidean distance can faithfully measure affinities between points that reside close to each other on the manifold. For remote points, however, the Euclidean distance is meaningless. In practice, we consider the estimated iRTFs as our acoustic features, which are merely an approximation of the RTFs. Consequently, the instantaneous features might lie beside the theoretical acoustic manifold, slightly violating the local linearity property of the manifold. However, it *can* be considered linear in the local neighborhoods for all practical purposes. This reasoning motivates us to apply a manifold learning technique that utilizes the local linearity property of the manifold, namely the LOCA framework.

#### B. Modified Local Conformal Autoencoder

In this work, we adopt the LOCA architecture proposed in [51] and suggest methodological modifications for its training process. Briefly, we are interested in an inaccessible latent manifold. We assume that there is an unknown measurement function

that relates latent states to their measurements in the high-dimensional observation space, and the measurement function is assumed to be smooth and nonlinear.

Our approach is based on an autoencoder architecture that aims to learn an embedding that is isometric to the latent low-dimensional manifold using its high-dimensional samples. To achieve this, a localized *burst sampling* strategy [57] is required. A burst is a collection comprising samples taken from a local neighborhood. Consequently, bursts can provide information on the *local* variability in the vicinity of each data point. In such a local neighborhood, the measurement function that maps the latent manifold to the observed high-dimensional space can be approximated as linear. This local linearity is described by the Jacobian of the measurement function, which captures how small changes in the latent space propagate to the observed space. As a result, within a sufficiently small neighborhood, Euclidean distances can be assumed to reliably reflect the true geometric structure of the manifold. That would enable “inverting” the nonlinear observation function using local approximations of the associated Jacobian.

We consider the case where the latent domain of our system of interest is a path-connected domain in the Euclidean space  $\mathcal{X} \subset \mathbb{R}^d$ . Observations of the system consist of samples captured by a measurement device, processed using (4), and are then given as a nonlinear smooth and bijective function  $f : \mathcal{X} \rightarrow \mathcal{Y}$ , where  $\mathcal{Y} \subset \mathbb{R}^D$  is the ambient or measurement space, and  $D \gg d$ . The high dimensional acoustic features, namely the RTFs, constitute the measurement space in our setting, while the speaker position is represented by states on a low dimensional manifold lying in the latent space. Assume we have  $N$  states in the latent domain,  $\{\mathbf{x}_i\}_{i=1}^N$ . If we wish to infer an embedding  $\rho : \mathbb{R}^D \rightarrow \mathbb{R}^d$ , such that the image of  $\rho \circ f$  is isometric to  $\mathcal{X}$ , then for each state,  $\mathbf{x}_i$ , we can look at a set of  $M$  adjacent latent states,  $\mathbf{X}_i = \{\mathbf{x}_i^{(j)}\}_{j=1}^M \in \mathbb{R}^D$ , such that  $\mathbf{X}_i \sim \mathcal{N}(\mathbf{x}_i, \sigma^2 \mathbf{I}_d)$  are independent random variables. Note that  $\rho$  is isometric to  $f^{-1}$ , meaning that they are the same up to some rotation, mirroring, and translation.

For a fixed  $1 \leq i \leq N$ , we define the burst (sample cloud)  $\mathbf{Y}_i = \{\mathbf{y}_i^{(j)}\}_{j=1}^M$  as the set of independent and identically distributed samples of the random variable  $f(\mathbf{X}_i) \in \mathbb{R}^D$ , where  $M$  denotes the number of samples in each burst. Following the mathematical derivation in [51], this definition requires that such embedded burst measurements be whitened. Namely,

$$\frac{1}{\sigma^2} \mathbf{C}(\rho(\mathbf{Y}_i)) = \mathbf{I} \quad \forall 1 \leq i \leq N, \quad (6)$$

where  $\mathbf{C}$  is the covariance matrix.

We use an encoder neural network ( $E$ ) to learn the embedding function  $\rho$ . Following (6), we define the loss term to be minimized at each iteration:

$$L_{\text{white}}(\rho) = \frac{1}{N} \sum_{i=1}^N \left\| \frac{1}{\sigma^2} \hat{\mathbf{C}}(\rho(\mathbf{Y}_i)) - \mathbf{I}_d \right\|_F^2, \quad (7)$$

with  $\rho$  the learned embedding function up to the current iteration and  $\hat{\mathbf{C}}(\rho(\mathbf{Y}_i))$  is the empirical covariance over a set of  $M$  realizations  $\{\rho(\mathbf{y}_i^{(j)})\}_{j=1}^M$ . Since  $f^{-1}$  is invertible, we further impose



an invertibility property on  $\rho$ . This effectively regularizes the solution of  $\rho$  by preventing noninvertible functions. Practically, invertibility is imposed using a decoder neural network ( $D$ ) parameterized by  $\gamma$ . This approach induces a loss term based on the mean squared error (MSE) between samples and their reconstructed counterparts:

$$L_{\text{recon}}(\rho, \gamma) = \frac{1}{N \cdot M} \sum_{i,j=1}^{N,M} \left\| \mathbf{y}_i^{(j)} - \gamma \left( \rho(\mathbf{y}_i^{(j)}) \right) \right\|_2^2. \quad (8)$$

We further add a new anchoring loss term to stabilize the solution and remove arbitrary scaling and rotations from the embedding learned by LOCA. Since we learn an embedding  $\rho$  that is only isometric to the inverse measurement function  $f^{-1}$ , we need to relate the embedded samples—i.e., the predicted speaker positions, to their real-world coordinates. In [52] and [51], the incorporation of real-world coordinates into the embedding was done only past the training stage, where an orthogonal transformation for the embedding was found by comparing the embedded anchor samples to their true positions. However, we found that using a small set of anchor points during training would be sufficient to accurately calibrate the embedding and improve the overall stability of the training process, as we demonstrate in Section IV-G. Assume that we have a set of  $N_L$  labeled samples,  $\{\mathbf{y}_k\}_{k=1}^{N_L}$ , whose latent states,  $\{\mathbf{x}_k\}_{k=1}^{N_L}$ , are known. We define the anchoring loss term as the mean squared error between the embedded samples and their latent states:

$$L_{\text{anchor}}(\rho) = \frac{1}{K} \sum_{k=1}^{N_L} \left\| \rho(\mathbf{y}_k) - \mathbf{x}_k \right\|^2. \quad (9)$$

Finally, our objective is to minimize the combined loss term

$$L = \lambda_{\text{white}} \cdot L_{\text{white}} + \lambda_{\text{recon}} \cdot L_{\text{recon}} + \lambda_{\text{anchor}} \cdot L_{\text{anchor}}, \quad (10)$$

where  $\lambda_{\text{white}}$ ,  $\lambda_{\text{recon}}$  and  $\lambda_{\text{anchor}}$  are the weighting factors for the whitening, the reconstruction, and the anchoring loss terms, respectively.

We note that in [51], the authors used a multi-step approach for iterative minimization of the reconstruction and whitening loss terms. In contrast, we suggest using a single-step approach to minimize the combined loss term (10), resulting in a much more stable learning process. This will also be further discussed and validated in Section IV-G. The training scheme for the LOCA framework is illustrated in Fig. 2.

### C. Synthetic Aperture Paradigm for Bursts Acquisition

Adopting the LOCA framework for our 1-dimensional problem,  $\{\mathbf{x}_i\}_i$ , the latent states in our system are simply the speaker DOAs,  $\{\theta(\ell)\}_\ell$ . The high-dimensional features we sample,  $\{\mathbf{y}_i\}_i$  are the iRTF features  $\{\hat{h}(l)\}_\ell$ .

One of the key differences in this work from [52], which involved performing a static 2-dimensional acoustic-scene mapping, is the reverse of roles between the sources and receivers. In [52], the sound sources were fixed, and the acoustic manifold was consequently controlled by the position of the microphone array. In that case, learning the manifold essentially resulted in

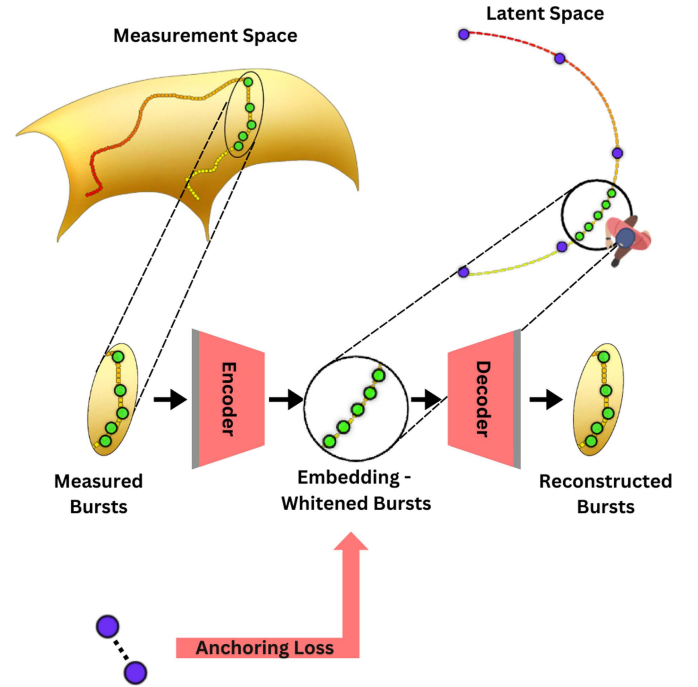


Fig. 2. An illustration of the modified LOCA framework training scheme. The latent space is a low-dimensional manifold in a high-dimensional measurement space. Bursts of adjacent samples from the measurement space are fed into the encoder, aiming to whiten their embedding (see (7)). The embeddings of the anchors (blue circles) are compared to their true positions via the anchoring loss (9). The embeddings are then fed into the decoder to reconstruct the input (8).

the mapping of the acoustic scene. In this work, the microphone array is fixed, the acoustic manifold is controlled by the position of the speakers, and the embedding we learn is used to track the DOA of the speaker. Importantly, training LOCA to infer an embedding that is isometric to the latent space requires clusters of neighboring samples (referred to as sample bursts). However, acquiring such data can be challenging. In [52], we devised a unique, albeit complex, 2-dimensional microphone array that allowed for the collection of bursts of adjacent samples. When we limit ourselves to the 1-dimensional setting, we can devise a simple solution that takes advantage of the dynamics of the speakers. When recording a moving speaker along the arc, we can think of measurements of consecutive instantaneous acoustic features as if they arrive from a local neighborhood in which the measurement function does not vary dramatically. This allows us to define local bursts. Our inspiration was the synthetic aperture paradigm in the radio detection and ranging (RADAR) domain [58], where movement is utilized to increase the resolution of the measured signal. In synthetic aperture RADAR (SAR), the motion of the RADAR antenna effectively increases its aperture, thereby improving its spatial resolution compared to conventional static radar systems. In our case, the goal is not to increase the resolution of the signal acquisition, but rather to enhance the spatial sampling of the acoustic manifold itself. Specifically, the movement of the speaker—rather than the movement of the microphones—facilitates a denser sampling of the acoustic manifold than would be achievable with stationary

speakers. Therefore, “synthetic aperture” in this context refers not to the virtual enlargement of the microphone array aperture, but to the synthetic accumulation of spatial samples as the speaker moves across a trajectory. This concept is analogous to SAR in spirit but with reversed roles: while SAR involves a moving sensor observing a stationary scene, our setup has stationary microphones observing a dynamic source.

In the following paragraphs, we will further formalize this concept, depicting our paradigm for collecting bursts of measurements.

Given the recording of a dynamic speaker confined to a 1-dimensional arc, our goal is to collect adjacent samples from the recording to train our network. The recording consists of two channels— $\{z_1(t), z_2(t)\}$ . We need to collect trustworthy samples to ensure the reliability of the extracted acoustic features. To estimate the iRTF features accurately, it is essential to use recording segments that have sufficient active speech. To ensure this, the first step is to apply a voice activity detector (VAD) algorithm that only maintains the active time frames. Here, we apply short time Fourier transform (STFT) using a window size of  $N_{\text{FFT}} = 2048$  and overlap of 75% between the adjacent windows to divide the recording into a group of time frames,  $\{t_i\}_i$ . Once the VAD algorithm is applied to the time frames, we are left only with the active time frames,

$$T = \{\ell_i \mid \text{VAD}(z_1(\ell_i)) = 1\}. \quad (11)$$

Define  $N_c^{\text{train}}$  as the number of context frames required to estimate the iRTFs during the training phase. Also, define  $N_c^{\text{test}}$  as the number of context frames needed for estimation during the test phase. Note that, in general, their value might differ, and it is up to the user to decide the number of context frames used during the test stage as it is closely related to the inference latency. Essentially, using more frames achieves better iRTF estimation but smears the DOA of the speaker. This phenomenon is especially pronounced at higher speaker velocities. An additional consideration when changing the inference latency is the mismatch between the training and test data. We aim to learn a model that can generalize its predictions to iRTF samples estimated with a different number of context frames. In Section IV-D we delve deeper into this matter.

To ensure that the estimation of training samples is done using sufficient active speech, we aim to keep only the time frames that are found in active time segments, i.e., their range of context frames is fully active:

$$\hat{T} = \left\{ \ell_i \in T \mid \forall \ell_i - \frac{N_c^{\text{train}}}{2} \leq \ell_j \leq \ell_i + \frac{N_c^{\text{train}}}{2}, \ell_j \in T \right\}. \quad (12)$$

After collecting all active time segments, our goal is to create adjacent bursts of iRTF samples. As discussed earlier, we can consider consecutive iRTF estimates as measurements that arrive from adjacent positions on the arc by utilizing the speaker’s movement. We set the size of each burst to  $M = 5$  and collect all groups of 5 time frames in  $\hat{T}$  with a spacing of 2 time-frames to construct our training set. The spacing taken between frames is a variable that determines the in-burst distance between iRTF

---

**Algorithm 1: Training-Bursts Acquisition.**


---

**Input:** Two-channel training recordings,  $\{z_1(t), z_2(t)\}$ , of dynamic speakers moving along a 1-dimensional arc.  
**Output:** Tensor of dimensions  $[N, M, D]$ , containing  $N$  bursts of  $M$  local iRTF samples, each of dimension  $D$ .  
**for** each train recording  $\{z_1(t), z_2(t)\}$  **do**  
 1) Apply a VAD to find which time-frames  $\ell_i$  contain active speech (see (11)).  
 2) Find the group  $\hat{T}$  which contains only the time frames  $\ell_i \in T$  whose entire set of context frames is active (see (12)).  
 3) Collect time-frame bursts - all groups of  $M = 5$  time-frames in  $\hat{T}$  with spacing of 2 time-frames (see (13)).  
 4) According to (4) and (5), estimate the  $D$ -dimensional iRTFs,  $\hat{h}(\ell)$ , for each time-frame  $\ell$  in each time-frame burst  $\tilde{T}_j$ . Store the iRTF-bursts in the output tensor.  
**end for**

---

measurements. Large spacing results in bursts of measurements that are not local, whereas spacing that is too small might not capture enough variability within the burst. Choosing a spacing of two frames is a good compromise between these two considerations. The time-frame bursts are of the form

$$\tilde{T}_i = \{\ell_i - 4, \ell_i - 2, \ell_i, \ell_i + 2, \ell_i + 4\}. \quad (13)$$

Finally, we use each of the collected time-frame bursts in  $\{\tilde{T}_i\}_i$  and estimate its iRTFs using (4) over each time frame. The DOAs at these time frames and their corresponding estimated iRTFs are, in fact, the latent states within each burst and their high-dimensional observations, respectively. Namely, we can write that  $\{x_i^{(j)}\}_{j=1}^M = \{\theta(\ell_i^{(j)})\}_{j=1}^M$  and  $\{y_i^{(j)}\}_{j=1}^M = \{\hat{h}(\ell_i^{(j)})\}_{j=1}^M$ , where each burst contains  $M = 5$  samples.

Since our primary focus is speech data, we consider only the frequency-bin range  $[41 - 320]$ , which corresponds to the frequencies 312–2500 Hz for our choice of  $N_{\text{FFT}} = 2048$ . This range includes most of the speech power distribution. Given that we are required to concatenate both the real and imaginary parts of the iRTF (5), we are left with features of dimension  $D = 560$ . Algorithm 1 outlines the process of training-bursts collection and is followed by Algorithm 2, which summarizes both the training process and the tracking methodology.

We note that the essential requirement for the sample bursts defined in Section III-B is that the latent states associated with each burst are independent and identically distributed (i.i.d.) Gaussian variables. In simpler terms, this means that the locally spatial distributions of samples within different bursts behave similarly and follow an isotropic Gaussian distribution. Note, however, that in the case where  $\sigma \ll 1$ , other isotropic distributions could also be used to model the local perturbations around states. To fulfill these requirements, we require that during the training stage, the speakers move at a constant velocity. We can then set the size of each burst to  $M = 5$  and gather five evenly spaced samples in time. It also follows that they are

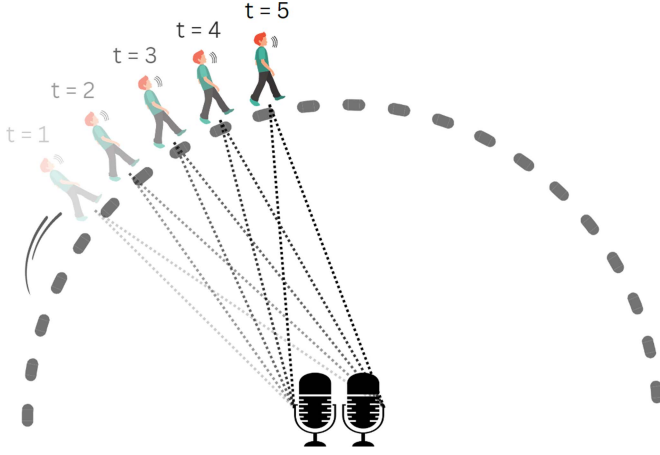


Fig. 3. An illustration of the synthetic aperture paradigm used for collecting bursts of measurements. Samples recorded at a group of consecutive time-frames,  $\tilde{T}_i$ , are grouped together to form the sample bursts  $\{y_i^{(j)}\}_{j=1}^M = \{\hat{h}(\ell_i^{(j)})\}_{j=1}^M$  for  $M = 5$ . For a speaker moving at a constant velocity, the latent states attributed to these samples,  $\{x_i^{(j)}\}_{j=1}^M = \{\theta(\ell_i^{(j)})\}_{j=1}^M$ , follow an isotropic 1-dimensional uniform distribution in the latent space, thus satisfying the requirements for using the LOCA framework.

---

**Algorithm 2:** Synthetic Aperture LOCA for Semi-Supervised Speaker Tracking.

---

**TRAINING STAGE**

**Input:**  $N_L$  labeled samples  $\{z_1^{\text{labeled}}(t), z_2^{\text{labeled}}(t)\}$ , synthetic aperture bursts tensor of dimensions  $[N, M, D]$ .

**Output:** An encoder network (E) parameterized by  $\rho$   
**for**  $N_{\text{epochs}}$  **do**

    Feed batches of bursts into the model, single-step minimizing the loss function in (10).

**end for**

**TEST STAGE - TRACKING**

**Input** A new recording containing a pair of signal measurements,  $\{z_1^{\text{test}}(t), z_2^{\text{test}}(t)\}$ , produced by an unknown source moving in an unknown location on the arc.

**Output** Estimated per frame DOA  $\hat{\theta}(\ell), \forall \ell$ .

- 1) According to (4) and (5) estimate the corresponding iRTF,  $\hat{h}(\ell)$  at time frame  $\ell$  using  $N_c^{\text{test}}$  context frames.
  - 2) Predict the DOA at time frame  $\ell$  using  $\hat{\theta}(\ell) = \rho(\hat{h}(\ell))$ .
- 

uniformly spaced in the latent space, thus satisfying the required conditions.

A conceptual depiction of this paradigm is illustrated in Fig. 3. We also add a block diagram in Fig. 4.

#### D. Model Training and Parameters

We implemented both the encoder and decoder using simple feed-forward architectures. The encoder consists of an initial layer of dimension  $D = 560$  with no nonlinearity, followed by five consecutive layers of size 200 with ‘Leaky-Relu’ activations, ending with a latent representation layer of size  $d = 1$ .

Similarly, the decoder consists of the latent representation layer with no nonlinearity, followed by four layers of size 200 with ‘tanh’ activations, an additional linear layer of size 200, and an output reconstruction layer of size  $D$ . We use batches of size 2048 samples and a learning rate of  $10^{-3}$ . The value of  $\sigma^2$  is set to  $6 \cdot 10^{-4}$ . In the original manuscript of LOCA, the authors apply alternating minimization steps between the whitening and reconstruction loss terms. We found, however, that a more stable solution is obtained by minimizing a combined loss function in (10). The loss terms are weighted using  $\lambda_{\text{white}} = 1$ ,  $\lambda_{\text{recon}} = 25$  and  $\lambda_{\text{anchor}} = 100$ . Finally, we use 90% of the learning-phase data during training, and the rest 10% are used for validation and determining the best model weights.

#### E. Future Extensions

Our work utilizes the LOCA framework for tracking a single speech source. Multi-source extensions may use speech sparsity, namely that speech sources are ‘isolated’ in the time-frequency (TF) domain [59], a property often referred to as W-disjoint orthogonality (WDO) [60]. This assumption was widely used for multi-source localization or TDOA estimation, e.g., [61], [62], [63]. In [39], the WDO assumption was used with the iRTF as an input feature to track multiple sources. In [37], a manifold-based multiple speaker tracking method was proposed. The manifold was inferred offline for a single and static source using a geometric method (diffusion maps), and the multi-speaker tracking was implemented through the application of the REM procedure with the iRTFs as input features. Extending our proposed LOCA-based DOA tracking to multiple speakers remains a challenge for future work.

### IV. EXPERIMENTAL STUDY

In this section, we conduct several experiments to compare the performance of our method with that of other schemes. We highlight the merits and limitations of our approach. Our experiments involve acoustic settings with reverberation levels of up to  $\text{RT}_{60} = 600$  ms, and we consider various signal to noise ratio (SNR) conditions of either 30, 20, 15, or 10 dB. The distance between the microphones is set to 8 cm, and the radius of the arc is set to 2 m. The evaluation and comparison of all tracking methods focus on the broadside of the microphone configuration, considering DOAs in the range  $[30^\circ - 150^\circ]$ .

In subsection IV-A, we specify the performance measures used to evaluate all approaches. In subsection IV-B, we compare the different approaches in a setup where the test data and the training data are drawn from the same constellation (position and orientation of the microphone array and the arc within the acoustic enclosure). Although the training and test stages share the same features, they differ in both velocity and speakers’ identities. Through subsections IV-C and IV-D, we assess the influence of two hyper-parameters on the performance of LOCA, also having matching acoustic setup between the training and test stage. Namely, the considered hyper-parameters are the number of speakers that provide anchors during training and the latency during test stage. In subsection IV-F, we test LOCA’s sensitivity to different array-speaker constellations during the test phase. In



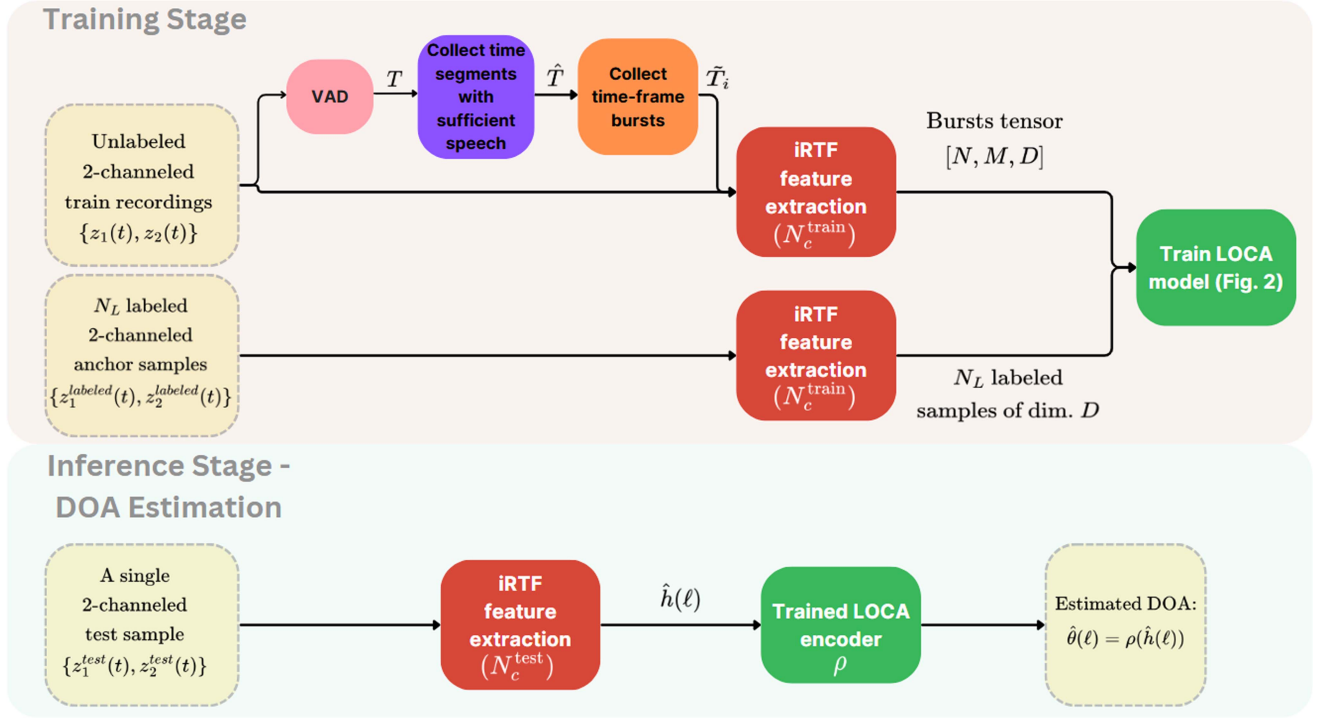


Fig. 4. Block diagram visualization of both the training (including the burst acquisition algorithm) and inference stages.

that case, we will have a mismatch between the training and test data. Finally, in subsection IV-G, we assess the influence of the two methodological changes we introduce for LOCA's training process. There, we focus on a specific acoustic setting where the test data matches the constellation of the training data.

#### A. Evaluation and Performance Measures

The performance of all methods was evaluated using two metrics. We define an outlier DOA estimate as an instance that diverges more than  $15^\circ$  from the true DOA. Therefore, our first performance measure is the Accuracy, which corresponds to the percentage of inlier estimates that do not exceed the  $15^\circ$  threshold. Denote:

$$N_{\text{inliers}} = \left| \left\{ i \mid 1 \leq i \leq N_{\text{test}}, \quad \left| \hat{\theta}_i - \theta_i \right| \leq 15^\circ \right\} \right| \quad (14)$$

The Accuracy is then defined as:

$$\text{Accuracy} = \frac{N_{\text{inliers}}}{N_{\text{test}}} \%. \quad (15)$$

The second performance measure is the root mean square error (RMSE). Only inlier estimates are considered for calculating the RMSE over the test set.

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{inliers}}} \sum_{i=1}^{N_{\text{inliers}}} (\hat{\theta}_i - \theta_i)^2}. \quad (16)$$

For each experiment, we will specify the test set used for the evaluation.

#### B. Simulation I - Method Comparison

In the following simulation, we compare our method with competing algorithms. As stated before, we present the results for different acoustic settings, each with a different level of noise and reverberation, examining different amounts of anchor samples. For each acoustic setting, a Monte Carlo simulation is conducted. A single simulation comprises 20 different constellations, which are determined by the position and orientation of the microphone array within a  $[6, 6, 2.4]$  m simulated room. In this simulation, for each constellation, a LOCA-neural network is trained and evaluated on the same constellation. Consequently, a distribution of the RMSEs over the different constellations and their average accuracy is obtained.

1) *Training Data*: To create the training dataset, 28 random speakers were sampled from the TIMIT dataset [64], equally distributed between males and females. Several recordings from each speaker were concatenated to create an extended recording of continuous speech. To simulate the moving speakers' signals, we used the signal generator package<sup>1</sup> for MATLAB. Each speaker was modeled as an omnidirectional source. For each constellation, the extended recordings of each speaker were used to simulate a single complete traversal over the arc at a constant velocity of 0.25 m/s.

The directional noise signals,  $u_i(t)$ , are generated using a simulated white Gaussian noise (WGN) source located at  $[4, 1, 1.8]$ . The source is located 60 cm above the plane where the microphones are stationed, at  $z = 1.2$  m. These sources are

<sup>1</sup> Available at the repository - [github.com/ehabets/Signal-Generator](https://github.com/ehabets/Signal-Generator)

static and are therefore generated by using the RIR generator<sup>2</sup> to simulate their RIRs, and then convolving the RIRs with WGN signals.

These 28 two-microphone signals of each constellation are then used along with Algorithm 1 to collect all possible bursts for training, resulting in 117,000 training bursts. The VAD algorithm that is used during this process is WebRTC-VAD,<sup>3</sup> with the aggressiveness level set to 2. The amount of context frames used to estimate the training iRTFs is  $N_c^{\text{train}} = 10$ .

In this simulation, we are exploring the scenario where only one of the speakers is used to provide labeled anchor samples. Specifically, we are investigating cases where the number of anchor samples  $N_L$  is 3, 4, or 5, and the anchors are evenly positioned along the arc.

An additional train dataset was created in order to evaluate LOCA's performance when it is trained using a dynamic WGN source signal traversing the arc only once. Here, a single train recording is simulated in the same manner as the TIMIT recordings. After applying Algorithm 1, we are left with 1,520 training-bursts only. The idea here is to exploit the fact that the power spectral density of the WGN is richer. This approach might be advantageous in simplifying the training process—requiring only a single traversal of the sound source over the arc, thus resulting in substantially fewer training bursts and much faster model convergence. The drawback, however, is that the learned model should be able to generalize to our test set, which consists of speech data. In the result section, the two approaches are distinguished by using either of the terms LOCA\_Speech or LOCA\_WGN.

2) *Test Data*: An additional 28 new speakers were sampled from the TIMIT dataset for the test dataset. For each constellation, the signals were simulated along the arc using the same approach as in the training dataset, with a single difference that the test speakers' velocities are selected from a set of values  $[0.25, 0.55, 0.85, 1.15]$  m/s. For each speaker, 20 test samples are drawn from the region of interest per test velocity, resulting in an overall of 2,240 test samples for each constellation. In this simulation, the number of context frames used to estimate the test iRTFs is  $N_c^{\text{test}} = 20$ .

3) *Competing Methods*: The competing methods we consider are deep ranking (DR) [49], manifold regularization learning (MRL) [46] and isometrically regularized variational auto encoder (IRVAE) [65]. It is essential to acknowledge that while both DR and MRL were developed and evaluated for the specific aim of speaker localization and tracking, IRVAE has not been applied to acoustic data thus far. A single challenge with IRVAE was that while it learns an isometric mapping for the embedding, the authors do not provide a means to anchor the embedding coordinates to the real-world positions. Our most effective solution involved post-training calibration of the embedding using the anchor set, similar to the calibration in [52].

4) *Results and Discussion*: Fig. 5 comprises multiple box plots that depict the performance of the different methods over the different acoustic settings, considering the various amounts

of anchor samples. Examining LOCA\_Speech, it is evident that we obtain very impressive results, having reached median RMSEs as low as  $4^\circ$  with an accuracy of nearly 100 %. Even though the training is performed on sources moving at a constant velocity, the results indicate that all methods can generalize their predictions to higher velocities as well. We stress that a minimal number of labeled anchor samples was required to achieve this notable performance. It is only under extreme conditions, where both the reverberation and the noise level are high, that we see a noticeable decrease in performance compared to the other methods. It may seem that sometimes other methods, namely the IRVAE, obtain better RMSEs. In most acoustic settings, however, LOCA\_Speech provides the best performance in terms of accuracy while, more importantly, judging the whiskers' length, it shows lower variance, thus providing more reliable convergence across different constellations. In that sense, IRVAE, which might appear to be a good approach, is inconsistent when trained on different constellations. In the case of LOCA\_WGN, we observe highly stable results, indicated by significantly thinner whiskers. However, the performance measures are not as robust as those achieved by LOCA\_Speech. This confirms our concern that the learned model might struggle to generalize from the WGN-induced training data to the dynamic-speaker-induced test data. Nevertheless, in cases where achieving fast model convergence is a top priority or when obtaining training speakers is challenging, a viable compromise would be to train on a single WGN recording.

We evaluate the computational resources required for the proposed method, comparing it to DR and IRVAE, the two other deep-learning-based approaches. The measures we consider are 1) Number of Model Parameters, 2) GPU Memory Usage,<sup>4</sup> 3) Inference Time,<sup>5</sup> 4) Peak Memory,<sup>6</sup> 5) CUDA Execution Time,<sup>7</sup> 6) FLOPS<sup>8</sup> - number of floating point operations per sample. The reported measures, which are reported in Table I, were all computed using NVIDIA's Tesla T4 GPU. While LOCA seems to reach better results, it is also the heaviest model of the compared approaches.

### C. Simulation II - Number of Speakers Providing Anchors

In the second simulation, we focus solely on LOCA\_Speech's performance. In the previous simulation, only a single speaker was used to generate a set of  $N_L$  uniformly spread labeled anchor samples on the arc. We aim to investigate the impact of the number of speakers providing labels on the results. In

<sup>4</sup>GPU memory usage is recorded using `nvidia-smi`, an NVIDIA command-line utility that tracks GPU resource consumption, including memory allocation. The reported memory usage is measured with a batch size of 1 for all methods to ensure consistency.

<sup>5</sup>Inference time is measured as the average time required for a single forward pass, computed over 1,000 runs on a CUDA-enabled device. Timing is recorded with `torch.cuda.Event(enable_timing=True).record()`.

<sup>6</sup>Peak memory consumption during the forward pass is tracked using `torch.cuda.memory.max_memory_allocated()`.

<sup>7</sup>CUDA execution time represents the average processing time per inference sample across 1,000 runs, measured using `torch.profiler`.

<sup>8</sup>FLOPS is the number of floating points operations per inference sample. It is estimated using PyTorch's `torch.utils.flop_counter.FlopCounterMode`.

<sup>2</sup>Available at the repository [github.com/ehabets/RIR-Generator](https://github.com/ehabets/RIR-Generator)

<sup>3</sup>Available at the repository - <https://github.com/wiseman/py-webrtcvad>



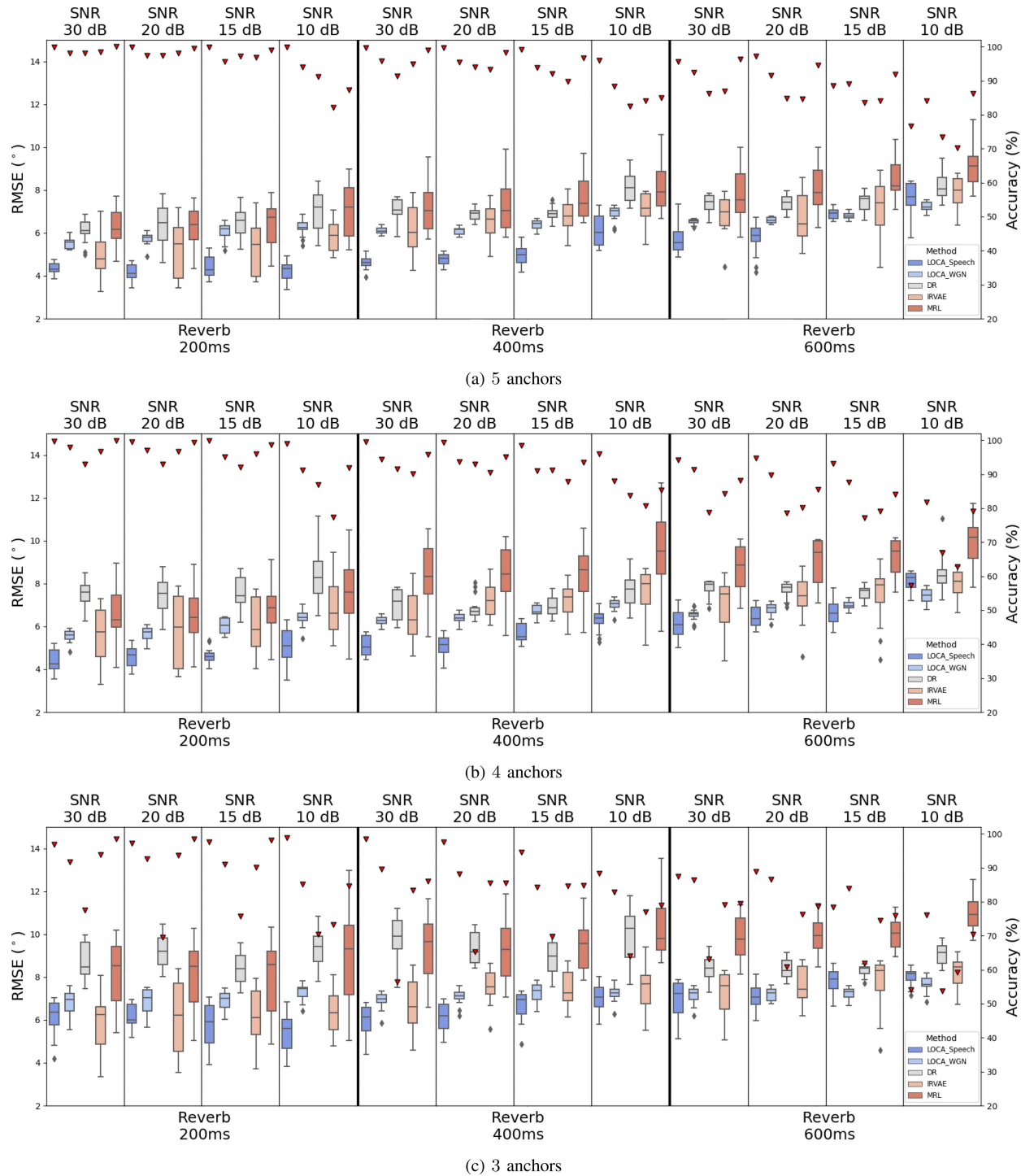


Fig. 5. A comparison between the RMSE and Accuracy measures for the different acoustic settings (Simulation I). Each sub-figure is attributed to a different amount of labeled anchor samples used. Within each sub-figure, a comparison is made for each possible combination of the examined SNR and reverberation levels. The boxes represent the distribution of RMSEs (16) achieved for 20 different constellations, while the red triangles denote their tracking accuracy (15).

this scenario, we fix  $N_L = 5$  uniformly spread labeled training samples per such speaker. Both the training and test data remain the same as in the previous simulation, with the only difference being the increase in number of anchored speakers.

1) *Results and Discussion:* The simulation results are depicted in Fig. 6. It is evident that while the improvement is

less pronounced at low to mid-reverberation levels, a significant enhancement is observed in highly reverberated acoustic environments. Specifically, notable improvement is observed when the number of anchored speakers is increased to three. This outcome effectively addresses the challenges we encountered earlier in the extreme acoustic conditions. While not requiring

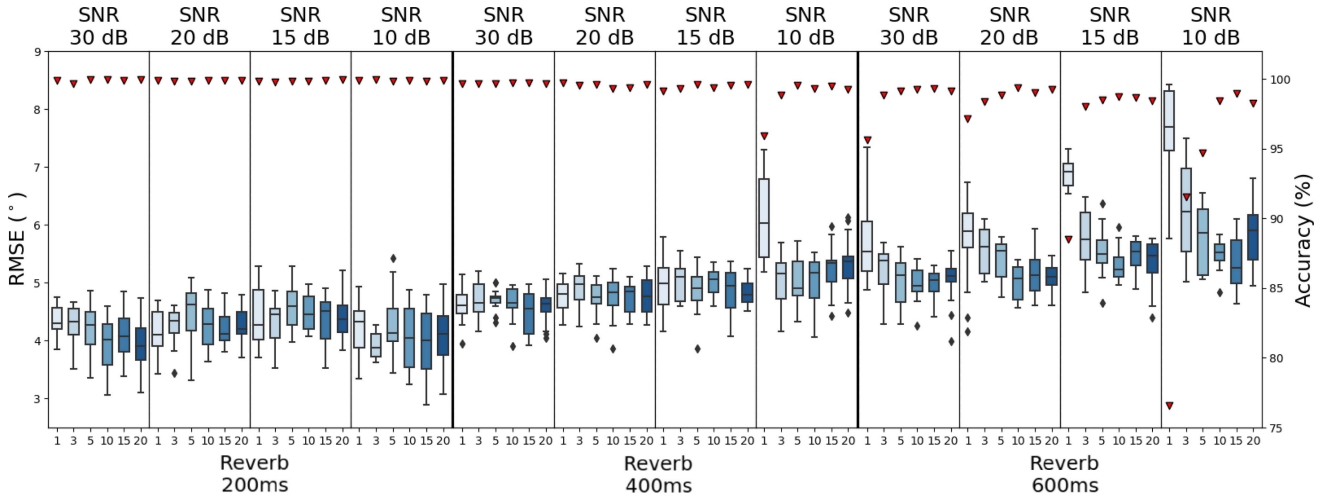


Fig. 6. Comparison between LOCA-Speech's performance over various amounts of speakers that provide labeled samples, demonstrated for the discussed reverberated and noisy conditions (Simulation II). For each speaker, 5 uniformly spaced anchor samples are drawn from the arc. The boxes represent the distribution of RMSEs (16) achieved for 20 different constellations, while the red triangles denote their tracking accuracy (15).

TABLE I  
COMPARISON OF THE REQUIRED COMPUTATIONAL RESOURCES FOR THE  
COMPETING METHODS

| Model                          | LOCA      | DR      | IRVAE   |
|--------------------------------|-----------|---------|---------|
| Parameters                     | 820,921   | 57,121  | 547,163 |
| GPU Memory Usage [MB]          | 1258      | 282     | 764     |
| Inference Time [ms]            | 0.93      | 0.31    | 0.51    |
| Peak Memory [MB]               | 40.85     | 27.7    | 29.6    |
| CUDA Execution Time [ $\mu$ s] | 249       | 53      | 109     |
| FLOPS                          | 1,636,000 | 114,020 | 544,800 |

a significant amount of additional labels, we are now able to consistently achieve good results in these settings both in terms of tracking accuracy and RMSE.

#### D. Simulation III - Influence of the Number of Context-Frames

When addressing the task of online dynamic speaker tracking, it is crucial to consider the proposed solution's ability to perform effectively across various latency levels. Consequently, we found interest in exploring LOCA's ability to generalize its predictions for different values of context frames utilized during the test phase compared to those used during the training phase. We implement the same training methodology used in Simulation I for this simulation. Similar to Simulation I, only a single anchored speaker is providing anchors for training. The primary difference lies in the evaluation of LOCA's performance across different number of context frames. These test datasets differ only in the number of context frames used for iRTF estimation, where the considered values for  $N_c^{test}$  range from 6 to 20.

1) *Results and Discussion:* Fig. 7 illustrates the evaluation results of LOCA on a different number of context frames. In most acoustic settings, a slight improvement is noticeable when

additional context frames are used to estimate the test iRTFs. This improvement is apparent both in Accuracy and in RMSE distribution, and it is due to the more precise estimation of the iRTF acoustic features when more context frames are utilized. However, a slight decrease in performance is noticeable in the case of  $N_c^{test} = 6$ , mainly in challenging reverberation and noise conditions. This performance degradation may be due to the decreased accuracy in the acoustic feature estimation as the number of context frames decreases, an issue that is especially evident in harsh environmental conditions. Nonetheless, it is apparent that LOCA presents robust generalization capabilities, offering significant flexibility in handling variations between the number of context frames during training and test phases.

#### E. Simulation IV - Generalization to Radial Distance Variations

As explained in detail in Section III-A, the only parameter governing the acoustic manifold in our setup is the speaker's DOA along a fixed-radius arc. This constraint allowed us to establish a direct mapping between the acoustic features and the single parameter that defines the manifold. In the current setup, we introduce a new degree of freedom during the test phase—the speaker's radial position—which becomes an additional parameter governing the manifold. To assess the model's ability to generalize under this new condition, we conducted an additional experiment where, during testing, the speaker's radius varied sinusoidally between 1.8 m and 2.2 m, while the model was still trained on a fixed 2 m arc. We followed the same Monte Carlo procedure, this time incorporating this more complex dynamics into the test data. The results, presented in Fig. 8, demonstrate that the model successfully adapts to these variations, maintaining robust performance even when the radial distance changes during inference. However, performance degradation is observed under highly reverberant conditions.

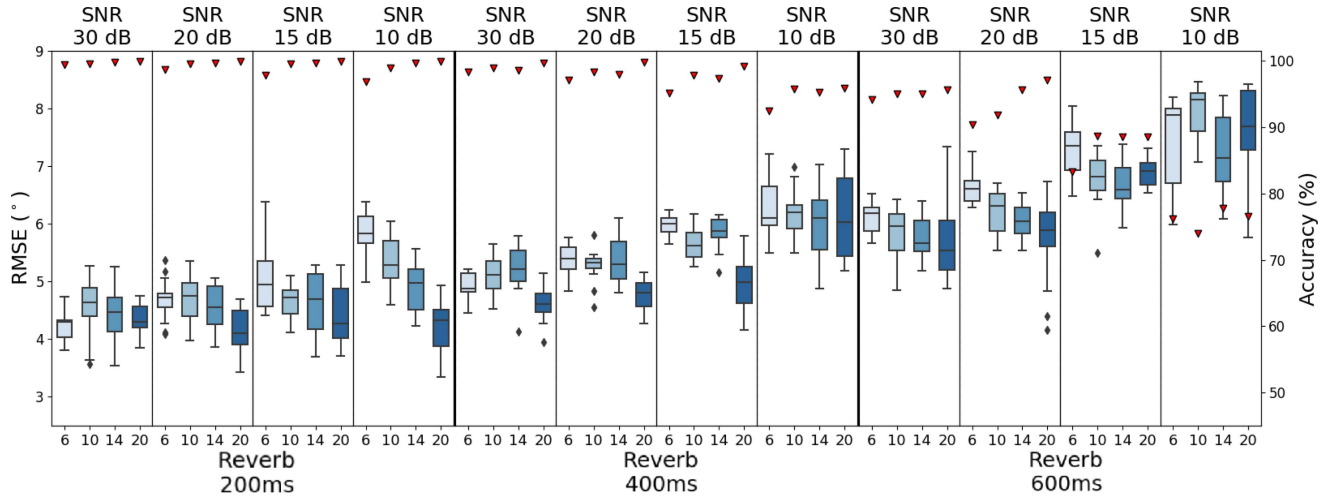


Fig. 7. Performance comparison of LOCA model that was trained on speech data (LOCA-Speech) when evaluated over a different number of test context frames  $N_c^{test}$  (Simulation III). Here, the amount of context frames used for training is  $N_c^{train} = 10$ .

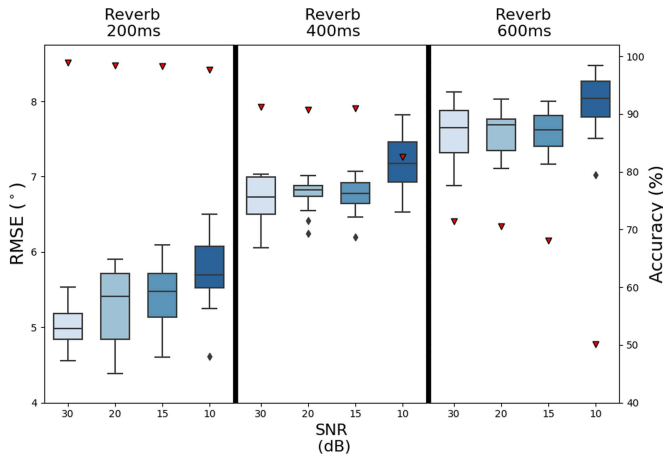


Fig. 8. Comparison between LOCA-Speech's performance for various reverberated and noisy conditions (Simulation IV), evaluated when the speaker moves along an arc with a sinusoidally varying radius between 1.8 m and 2.2 m. The results demonstrate the model's ability to generalize to radial distance variations during inference.

#### F. Simulation V - Sensitivity to Array-Speaker Constellation

In our final simulation, we aim to investigate LOCA's sensitivity to different array-speaker constellations. Essentially, this could be thought of as some sort of spatial generalization interpolation of a trained model to new mic-array positions and orientations within the enclosure. Given an acoustic setting, LOCA is trained on dynamic speakers drawn from a set of training constellations. The task is then to track speakers drawn from new constellations in the enclosure that were not seen during training. Recall that different constellations are simply determined by the position and orientation of the microphone pair and the speakers within the enclosure. We note that all of the constellations in this simulation are constrained to the plane  $z = 1.2$  m. Furthermore, the constellations under consideration involve speakers positioned at a radius of  $r = 2$  m from the microphone pairs. The position of the noise source remains

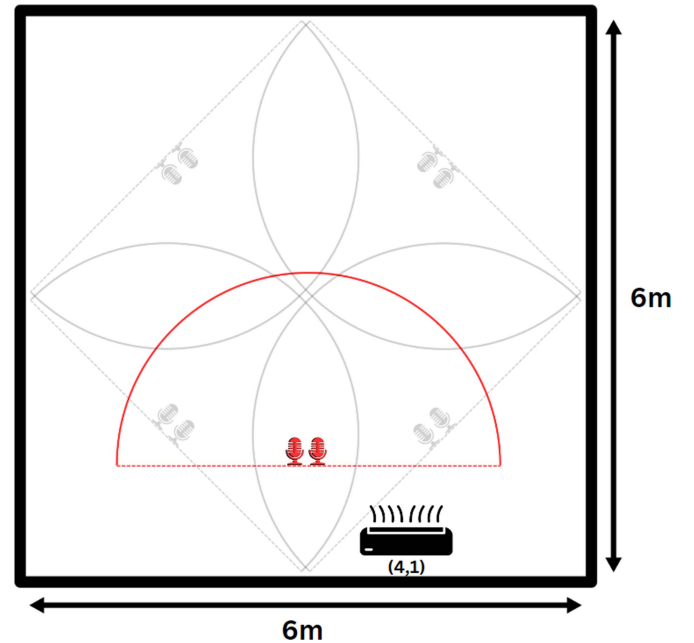


Fig. 9. An abstract visualization for Simulation V - assessing LOCA's sensitivity to array-speaker constellation. Our training data consists of speakers traveling along arcs of a number of constellations (gray arcs). The test data belongs to constellations not seen during training (red arcs). The interference source is located on  $[x, y] = [4, 1]$ , similar to the previous simulations. Both training and evaluation are conducted under the same acoustic setting (i.e., under the same reverberation and SNR levels).

unaltered. Since this scenario is significantly more challenging than previous ones, we excluded the 10 dB SNR case, as the method did not produce meaningful results under such severe noise conditions. The above idea is conceptualized in Fig. 9.

1) *Training Data:* To generate the training dataset, we considered 24 different training constellations for any given acoustic setting. For ideal training, different training constellations are chosen so that the entire room is covered in the most symmetrical manner. Each training constellation includes 7 random speakers



drawn from the TIMIT dataset. Similarly to Simulation I, we use the extended speaker recordings and the signal generator package for MATLAB. The speakers were simulated with a constant velocity of 0.25 m/s, this time while conducting back-and-forth traversal along the arc. After applying Algorithm 1 for training-bursts collection with  $N_c^{\text{train}} = 10$  training context frames, each training constellation provided approximately 9,100 bursts, resulting in a total of 218,400 bursts from the entire acoustic enclosure. For this simulation, which is already very challenging, we allow each of the 7 speakers to provide 5 equally-spaced anchor samples in each training constellation.

2) *Test Data:* For evaluation, we consider 24 test constellations that the model has not seen during training. In each constellation, 10 speakers are considered for evaluation. Once again, we are interested in evaluating the model's performance over multiple velocities, examining speaker velocities of [0.25, 0.55, 0.85, 1.15] m/s. For each speaker, the iRTF features are estimated using  $N_c^{\text{test}} = 20$ , of which 20 samples are drawn per each of the four speaker velocities. This results in a total of 19,200 test samples per acoustic setting.

3) *Results and Discussion:* In this simulation, it is important to note that the training is conducted collectively using data extracted from multiple constellations within the enclosure. This differs from our previous simulations, in which the models were trained and evaluated individually on the exact constellations for each acoustic setting. In the current setup, both the training and test datasets are drawn from all possible array-speaker constellations in the enclosure. Consequently, we train a single model for the entire enclosure. Unlike previously, the results are not visualized through box plots of a Monte Carlo simulation but are instead represented in a tabular format, showcasing the performance measures achieved across the combined test dataset. The baseline model used for comparison is the IRVAE discussed earlier. The MRL approach was not considered for the task, as the great number of training samples is an obstacle for kernel-based methods. Additionally, multiple attempts to utilize the DR model for the task approach were not successful at all.

Table II presents the simulation results. Aside from showing the results of the current experiment (Cur.), we include the nominal results (Nom.) from Simulation I to show the degradation in performance due to dealing with the more challenging setup. It is worth noting that both IRVAE and LOCA demonstrate considerable success in this challenging task, LOCA appears less sensitive to the array-speaker constellation. As observed, both approaches generally experience a decline in performance as the noise and reverberation levels increase. Inspecting LOCA, we see that, as expected, both the prediction accuracy and the prediction RMSE are negatively affected in this setup compared to previous simulations. However, in almost every acoustic setting, LOCA outperforms IRVAE when comparing both median RMSE and accuracy measures. A single acoustic setting that sets the most significant obstacle for LOCA is the challenging setting of SNR = 15 dB and  $RT_{60} = 600$  ms. Under these conditions, the model learns an unreliable embedding for the manifold, and consequently, predictions significantly diverge from the actual positions.

TABLE II  
COMPARISON BETWEEN THE MODIFIED LOCA AND IRVAE IN THE  
ARRAY-SPEAKER SENSITIVITY EXPERIMENT (SIMULATION V))

| Model | SNR<br>(dB) |      | 200<br>(ms)     | 400<br>(ms)     | 600<br>(ms)     |
|-------|-------------|------|-----------------|-----------------|-----------------|
| LOCA  | 30          | Cur. | 5.9 (95)        | <b>7.0 (90)</b> | <b>7.4 (84)</b> |
|       |             | Nom. | 4.3 (100)       | 4.6 (100)       | 5.5 (96)        |
|       | 20          | Cur. | <b>5.9 (97)</b> | <b>7.6 (86)</b> | <b>7.2 (86)</b> |
|       |             | Nom. | 4.1 (100)       | 4.8 (100)       | 5.9 (97)        |
|       | 15          | Cur. | <b>5.8 (98)</b> | <b>7.3 (82)</b> | 7.7 (52)        |
|       |             | Nom. | 4.3 (100)       | 5.0 (99)        | 6.9 (89)        |
| IRVAE | 30          | Cur. | <b>5.0 (97)</b> | 8.9 (71)        | 7.6 (84)        |
|       |             | Nom. | 4.8 (99)        | 6.0 (95)        | 7.0 (87)        |
|       | 20          | Cur. | 7.3 (93)        | 7.5 (83)        | 7.5 (84)        |
|       |             | Nom. | 5.5 (98)        | 6.6 (93)        | 6.4 (85)        |
|       | 15          | Cur. | 6.8 (88)        | 7.6 (82)        | <b>7.6 (76)</b> |
|       |             | Nom. | 5.5 (97)        | 6.8 (90)        | 7.4 (84)        |

The results are represented for the current (Cur.) experiment in terms of median RMSE in cm, and accuracy percentage (in brackets). The nominal (Nom.) results are also included, showcasing the methods' performance in Simulation I.

### G. Ablation Study

As stressed earlier in this manuscript, we have included several methodological adjustments to the LOCA training framework to achieve more consistent convergence. In [51], a post-training calibration was suggested for incorporating the real-world coordinates into the learned embedding. Later, in [52], the single-step minimization of the loss function was suggested while the post-training calibration approach was still used. In the current work, we also include the single-step minimization approach as an adjustment for the loss term, which is an alternative to the multi-step minimization approach (Section III-D). However, the *main* modification we introduce involves incorporating the additional anchoring loss term into the optimization process.

In the following brief ablation study, we demonstrate the training stability of our proposed approach by comparing it with two ablated versions. The first ablated version, referred to as 'Multi-step minimization,' is identical to our proposed approach except that it integrates multi-step minimization of the loss function. The second ablated approach, referred to as 'Post-training calibration,' also retains the core components of our approach but includes post-training calibration of the embedding instead of the proposed additional anchoring loss term. By comparing the performance of these ablated approaches with our proposed approach, we aim to assess each component's impact on the LOCA framework's overall effectiveness.

The models are trained and evaluated using our 1-dimensional data, following the performance measures discussed earlier. We conduct a Monte Carlo simulation for the task, mirroring the approach used in Simulation I and utilizing the same training and test datasets. Specifically, we consider an acoustic setting with reverberation and noise levels of  $RT_{60} = 600$  ms and 30 dB, respectively. The results are presented in Fig. 10. It is evident that the proposed approach achieves both the

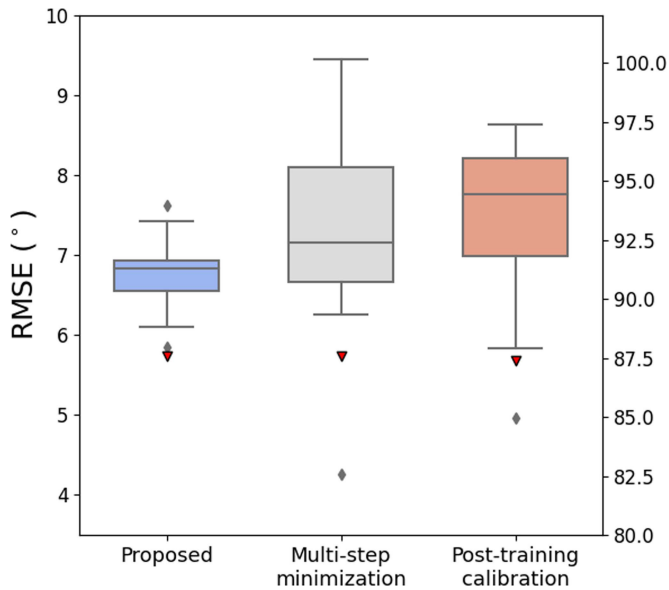


Fig. 10. An ablation study on the stability of training for the different training approaches of LOCA. The results shown are attributed to the acoustic setting of  $RT_{60} = 600$  ms and 30 dB SNR.

lowest median RMSE and the narrowest whiskers. In comparison, the ablated versions achieve slightly inferior results in median RMSE, with significantly wider and skewed RMSE distributions observed across the 20 constellations in the enclosure. These findings clearly indicate that the proposed approach leads to significantly more stable and reliable overall performance.

## V. CONCLUSION

In this paper, we adopt LOCA, a DNN-based manifold learning approach. LOCA, which was proven viable for the task of acoustic scene mapping, is now used for semi-supervised dynamic speaker tracking. We introduce two novel adjustments to the original algorithm to further optimize the training process: the anchoring loss term and the single-step minimization. By utilizing high-dimensional instantaneous relative transfer function (iRTF) acoustic features lying on a manifold, the model learns an embedding that is isometric to the latent manifold underlying the data. Our approach aims to utilize a minimal amount of labeled data, a resource often scarce. To achieve that, we apply the synthetic aperture training paradigm to exploit local relations between subsequent samples, effectively leveraging the speakers' movement during the training phase. A series of simulations showcases the proposed method's ability to track speakers under challenging noise and reverberation conditions and the influence of the number of labeled samples. Furthermore, it demonstrates notable generalization capabilities, enabling predictions for both higher speaker velocities, varying levels of latency, and varying radial distance from the microphone array. Lastly, it shows robustness for different array-speaker constellations within the acoustic enclosure.

## REFERENCES

- [1] B. H. Yoshimi and G. S. Pingali, "A multimodal speaker detection and tracking system for teleconferencing," in *Proc. 10th ACM Int. Conf. Multimedia*, ser. MM02, Dec. 2002, pp. 427–428.
- [2] H. Gamper, S. Tervo, and T. Lokki, "Speaker tracking for teleconferencing via binaural headset microphones," in *Proc. Int. Workshop Acoust. Signal Enhancement*, 2012, pp. 1–4.
- [3] C. Zieger, A. Brutti, and P. Svaizer, "Acoustic based surveillance system for intrusion detection," in *Proc. 6th IEEE Int. Conf. Adv. Video Signal Based Surveill.*, Sep. 2009, pp. 314–319.
- [4] M. Crocco, M. Cristani, A. Trucco, and V. Murino, "Audio surveillance: A systematic review," *ACM Comput. Surv.*, vol. 48, no. 4, pp. 1–46, 2016.
- [5] C. Knapp and G. Carter, "The generalized correlation method for estimation of time delay," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 24, no. 4, pp. 320–327, Aug. 1976.
- [6] H. Do, H. F. Silverman, and Y. Yu, "A real-time SRP-PHAT source location implementation using stochastic region contraction (SRC) on a large-aperture microphone array," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2007, pp. I-121–I-124.
- [7] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Trans. Antennas Propag.*, vol. 34, no. 3, pp. 276–280, Mar. 1986.
- [8] J. H. DiBiase, H. F. Silverman, and M. S. Brandstein, "Robust localization in reverberant rooms," in *Microphone Arrays*. Berlin, Germany: Springer, 2001, pp. 157–180.
- [9] J. P. Dmochowski and J. Benesty, "Steered beamforming approaches for acoustic source localization," in *Speech Processing in Modern Communication*. Berlin, Germany: Springer, 2010, pp. 307–337.
- [10] T. G. Dvorkind and S. Gannot, "Time difference of arrival estimation of speech source in a noisy and reverberant environment," *Signal Process.*, vol. 85, no. 1, pp. 177–204, Jan. 2005.
- [11] A. Levy, S. Gannot, and E. A. P. Habets, "Multiple-hypothesis extended particle filter for acoustic source localization in reverberant environments," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 19, no. 6, pp. 1540–1555, Aug. 2011.
- [12] S. Chakrabarty, K. Kowalczyk, M. Taseska, and E. A. P. Habets, "Extended kalman filter with probabilistic data association for multiple non-concurrent speaker localization in reverberant environments," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2014, pp. 7445–7449.
- [13] C. Evers, E. A. P. Habets, S. Gannot, and P. A. Naylor, "DoA reliability for distributed acoustic tracking," *IEEE Signal Process. Lett.*, vol. 25, no. 9, pp. 1320–1324, Sep. 2018.
- [14] D. Bechler, M. Grimm, and K. Kroschel, "Speaker tracking with a microphone array using Kalman filtering," *Adv. Radio Sci.*, vol. 1, pp. 113–117, May 2003.
- [15] D. Salvati, C. Drioli, and G. L. Foresti, "Localization and tracking of an acoustic source using a diagonal unloading beamforming and a Kalman filter," in *Proc. LOCATA Chall. Workshop Satellite Event IWAENC*, 2018.
- [16] S. Gannot and T. G. Dvorkind, "Microphone array speaker localizers using spatial-temporal information," *EURASIP J. Adv. Signal Process.*, vol. 2006, no. 1, May 2006, Art. no. 059625.
- [17] U. Klee, T. Gehrig, and J. McDonough, "Kalman filters for time delay of arrival-based source localization," *EURASIP J. Adv. Signal Process.*, vol. 2006, no. 1, Mar. 2006, Art. no. 012378.
- [18] Z. Liang, X. Ma, and X. Dai, "Robust tracking of moving sound source using multiple model Kalman filter," *Appl. Acoust.*, vol. 69, no. 12, pp. 1350–1355, Dec. 2008.
- [19] J. Traa and P. Smaragdis, "A wrapped Kalman filter for azimuthal speaker tracking," *IEEE Signal Process. Lett.*, vol. 20, no. 12, pp. 1257–1260, Dec. 2013.
- [20] J. Vermaak and A. Blake, "Nonlinear filtering for speaker tracking in noisy and reverberant environments," in *Proc. IEEE Int. Conf. Acoustics, Speech, Signal Process.*, 2001, pp. 3021–3024.
- [21] D. Ward, E. Lehmann, and R. Williamson, "Particle filtering algorithms for tracking an acoustic source in a reverberant environment," *IEEE Speech Audio Process.*, vol. 11, no. 6, pp. 826–836, Nov. 2003.
- [22] Y. Chen and Y. Rui, "Real-time speaker tracking using particle filter sensor fusion," *Proc. IEEE*, vol. 92, no. 3, pp. 485–494, Mar. 2004.
- [23] X. Zhong and J. R. Hopgood, "Nonconcurrent multiple speakers tracking based on extended Kalman particle filter," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Mar. 2008, pp. 293–296.
- [24] T. Li and W. Ser, "Three dimensional acoustic source localization and tracking using statistically weighted hybrid particle filtering algorithm," *Signal Process.*, vol. 90, no. 5, pp. 1700–1719, May 2010.

- [25] N. Roman and D. Wang, "Binaural tracking of multiple moving sources," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 16, no. 4, pp. 728–739, May 2008.
- [26] K. Weisberg and S. Gannot, "Multiple speaker tracking using coupled HMM in the STFT domain," in *Proc. IEEE Int. Workshop Comput. Adv. Multi-Sensor Adaptive Process.*, Dec. 2019, pp. 286–290.
- [27] N. T. Pham, W. Huang, and S. H. Ong, "Multiple sensor multiple object tracking with GMPHD filter," in *Proc. 10th IEEE Int. Conf. Inf. Fusion*, 2007, pp. 1–7.
- [28] M. Beard and S. Arulampalam, "Performance of PHD and CPHD filtering versus JIPDA for bearings-only multi-target tracking," in *Proc. 15th IEEE Int. Conf. Inf. Fusion*, 2012, pp. 542–549.
- [29] Q. Liu, W. Wang, T. de Campos, P. J. B. Jackson, and A. Hilton, "Multiple speaker tracking in spatial audio via PHD filtering and depth-audio fusion," *IEEE Trans. Multimedia*, vol. 20, no. 7, pp. 1767–1780, Jul. 2018.
- [30] Z. Fu, F. Angelini, J. Chambers, and S. M. Naqvi, "Multi-level cooperative fusion of GM-PHD filters for online multiple human tracking," *IEEE Trans. Multimedia*, vol. 21, no. 9, pp. 2277–2291, Sep. 2019.
- [31] K. Weisberg, B. Laufer-Goldshtein, and S. Gannot, "Simultaneous tracking and separation of multiple sources using factor graph model," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 2848–2864, 2020.
- [32] O. Schwartz and S. Gannot, "Speaker tracking using recursive EM algorithms," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 22, no. 2, pp. 392–402, Feb. 2014.
- [33] D. Giacobello, "An online expectation-maximization algorithm for tracking acoustic sources in multi-microphone devices during music playback," in *Proc. 26th Eur. Signal Process. Conf.*, 2018, pp. 1547–1551.
- [34] Y. Dorfan, A. Plinge, G. Hazan, and S. Gannot, "Distributed expectation-maximization algorithm for speaker localization in reverberant environments," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 26, no. 3, pp. 682–695, Mar. 2018.
- [35] X. Li, Y. Ban, L. Girin, X. Alameda-Pineda, and R. Horaud, "Online localization and tracking of multiple moving speakers in reverberant environments," *IEEE J. Sel. Topics Signal Process.*, vol. 13, no. 1, pp. 88–103, Mar. 2019.
- [36] Y. Dorfan, B. Schwartz, and S. Gannot, "Forward-backward recursive expectation-maximization for concurrent speaker tracking," *EURASIP J. Audio, Speech, Music Process.*, vol. 2021, no. 1, Jan. 2021, Art. no. 2.
- [37] A. Bross and S. Gannot, "Training-based multiple source tracking using manifold-learning and recursive expectation-maximization," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 1124–1140, Mar. 2023.
- [38] S. Chakrabarty and E. A. Habets, "Multi-speaker DOA estimation using deep convolutional networks trained with noise signals," *IEEE J. Sel. Topics Signal Process.*, vol. 13, no. 1, pp. 8–21, Mar. 2019.
- [39] H. Hammer, S. E. Chazan, J. Goldberger, and S. Gannot, "Dynamically localizing multiple speakers based on the time-frequency domain," *EURASIP J. Audio, Speech, Music Process.*, vol. 2021, no. 1, Apr. 2021, Art. no. 16.
- [40] A. Bohlender, A. Spriet, W. Tirry, and N. Madhu, "Exploiting temporal context in CNN based multisource DOA estimation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 1594–1608, 2021.
- [41] S. Adavanne, A. Politis, and T. Virtanen, "Localization, detection and tracking of multiple moving sound sources with a convolutional recurrent neural network," in *Proc. Acoust. Scenes Events Workshop*, 2019, pp. 20–24.
- [42] A. Bohlender, L. Roelens, and N. Madhu, "Improved deep speaker localization and tracking: Revised training paradigm and controlled latency," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Jun. 2023, pp. 1–5.
- [43] J. Datta, M. Adams, and C. Perez, "Dense-u-Net assisted localization of speech sources in motion under reverberant conditions," in *Proc. 12th IEEE Int. Conf. Control Automat. Inf. Sci.*, 2023, pp. 440–447.
- [44] A. Schwartz, E. Hadad, S. Gannot, and S. E. Chazan, "Array configuration mismatch in deep DOA estimation: Towards robust training," in *Proc. IEEE Workshop Appl. Signal Process. to Audio Acoust.*, New Paltz, USA, Oct. 2023, pp. 1–5.
- [45] B. Laufer-Goldshtein, R. Talmon, and S. Gannot, "Semi-supervised source localization on multiple manifolds with distributed microphones," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 7, pp. 1477–1491, Jul. 2017, doi: [10.1109/TASLP.2017.2696310](https://doi.org/10.1109/TASLP.2017.2696310).
- [46] B. Laufer-Goldshtein, R. Talmon, and S. Gannot, "A hybrid approach for speaker tracking based on TDOA and data-driven models," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 26, no. 4, pp. 725–735, Apr. 2018.
- [47] M. J. Bianco, S. Gannot, E. Fernandez-Grande, and P. Gerstoft, "Semi-supervised source localization in reverberant environments with deep generative modeling," *IEEE Access*, vol. 9, pp. 84956–84970, 2021.
- [48] R. Opoichinsky, B. Laufer-Goldshtein, S. Gannot, and G. Chechik, "Deep ranking-based sound source localization," in *Proc. IEEE Workshop Appl. Signal Process. Audio Acoust.*, Oct. 2019, pp. 283–287.
- [49] R. Opoichinsky, G. Chechik, and S. Gannot, "Deep ranking-based DOA tracking algorithm," in *Proc. 29th Eur. Signal Process. Conf.*, 2021, pp. 1020–1024.
- [50] P.-A. Grumiaux, S. Kitić, L. Girin, and A. Guérin, "A survey of sound source localization with deep learning methods," *J. Acoustical Soc. America*, vol. 152, no. 1, pp. 107–151, 2022.
- [51] E. Peterfreund et al., "Local conformal autoencoder for standardized data coordinates," *Proc. Nat. Acad. Sci.*, vol. 117, no. 49, pp. 30918–30927, Nov. 2020.
- [52] I. Cohen, S. Gannot, and O. Lindenbaum, "Unsupervised acoustic scene mapping based on acoustic features and dimensionality reduction," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2024, pp. 386–390.
- [53] B. Laufer-Goldshtein, R. Talmon, and S. Gannot, "Semi-supervised sound source localization based on manifold regularization," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 24, no. 8, pp. 1393–1407, Aug. 2016.
- [54] B. Laufer-Goldshtein, R. Talmon, and S. Gannot, "A Study on Manifolds of Acoustic Responses," in *Latent Variable Analysis and Signal Separation*. Berlin, Germany: Springer Int. Publishing, 2015, pp. 203–210.
- [55] B. Laufer-Goldshtein, R. Talmon, and S. Gannot, "Data-driven multi-microphone speaker localization on manifolds," *Foundations Trends Signal Process.*, vol. 14, no. 12, pp. 1–161, 2020.
- [56] J. B. Tenenbaum, V. de Silva, and J. C. Langford, "A global geometric framework for nonlinear dimensionality reduction," *Science*, vol. 290, no. 5500, pp. 2319–2323, Dec. 2000.
- [57] A. Singer and R. R. Coifman, "Non-linear independent component analysis with diffusion maps," *Appl. Comput. Harmon. Anal.*, vol. 25, no. 2, pp. 226–239, Sep. 2008.
- [58] W. M. Brown, "Synthetic aperture radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-3, no. 2, pp. 217–229, Mar. 1967.
- [59] Y. Deville and M. Puigt, "Temporal and time-frequency correlation-based blind source separation methods. part I: Determined and underdetermined linear instantaneous mixtures," *Signal Process.*, vol. 87, no. 3, pp. 374–407, 2007.
- [60] S. Rickard and O. Yilmaz, "On the approximate W-disjoint orthogonality of speech," in *Proc. IEEE Int. Conf. Acoustics, Speech, Signal Process.*, 2002, pp. I-529–I-532.
- [61] S. Arberet, R. Gribonval, and F. Bimbot, "A robust method to count and locate audio sources in a multichannel underdetermined mixture," *IEEE Trans. Signal Process.*, vol. 58, no. 1, pp. 121–133, Jan. 2010.
- [62] C. Blandin, A. Ozerov, and E. Vincent, "Multi-source TDOA estimation in reverberant audio using angular spectra and clustering," *Signal Process.*, vol. 92, no. 8, pp. 1950–1960, 2012.
- [63] D. Pavlidis, A. Griffin, M. Puigt, and A. Mouchtaris, "Real-time multiple sound source localization and counting using a circular microphone array," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 21, no. 10, pp. 2193–2206, Oct. 2013.
- [64] J. Garofolo et al., "Darpa timit acoustic-phonetic continuous speech corpus CD-ROM," NIST Interagency/Internal Rep. (NISTIR), Nat. Inst. Stand. Technol., Gaithersburg, MD, USA, 1993.
- [65] Y. Lee, S. Yoon, M. Son, and F. C. Park, "Regularized autoencoders for isometric representation learning," in *Proc. Int. Conf. Learn. Representations*, 2022.